TEXORA · LECTURE SHEET

Computer Networks and the Internet

What a network is and why we build one, the main network types and topologies, the hardware and media that carry data, and how the Internet, its addresses, names and protocols hold together in daily use.

CSE-2105 · FUNDAMENTALS OF COMPUTING · TOPIC 4 · LEVEL 2 TERM 1

1. What a computer network is

A **computer network** is a collection of two or more computers and other devices, connected by wired or wireless links, so that they can exchange data and share resources. Each connected machine is called a **node** or **host**. A single computer working alone is a standalone machine; the moment it can send a file, share a printer or reach a shared database with another machine, it has become part of a network.

Networks are built for practical benefits, not for their own sake:

- **Resource sharing.** One printer, one scanner, one large storage drive or one Internet connection can serve many users, instead of buying a separate set for every desk.
- **File and information sharing.** A file stored in one shared place is available to everyone who is allowed to see it, in its latest version, without copying it onto memory sticks.
- **Communication.** Email, messaging, voice calls and video meetings all move over networks, usually faster and cheaper than the alternatives they replaced.
- **Central control and backup.** Software, user accounts and security settings can be managed from one server, and important data can be backed up in one routine operation.
- **Reliability.** If one machine fails, work can often continue on another, because the data and the services do not live on only one computer.

The costs are real and should be stated plainly: networking hardware and cabling cost money, a network needs skilled management, and a connected machine can be attacked through the connection. Good design weighs the benefit against these risks from the start.

2. Types of network by size

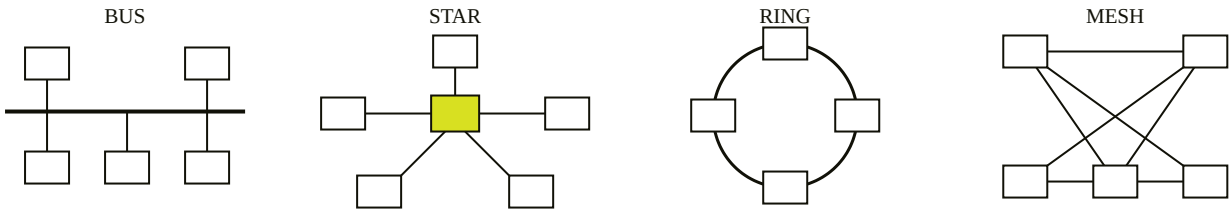
The most common classification is by geographical span. The three names every student must be able to compare are LAN, MAN and WAN.

TYPE	SPAN	TYPICAL EXAMPLE	NOTES
LAN — Local Area Network	A room, a building or a small campus, up to a few kilometres	Computers in a university laboratory, an office or a home	Usually owned by one organisation; high speed; low error rate; uses its own cabling or Wi-Fi
MAN — Metropolitan Area Network	A city or a large town, roughly 5 to 50 km	A cable television network, or several campuses of one university linked across a city	Larger than a LAN, smaller than a WAN; often built on fibre links provided by a carrier
WAN — Wide Area Network	A country, a continent or the whole world	The Internet; the linked branches of a bank in many districts	Built from many LANs joined by public or leased lines; slower per link and owned by several parties

A useful memory rule: a LAN is *yours*, in *your building*; a WAN is built from links that belong to *other people* as well. A MAN sits between the two, at city scale. The Internet itself is best described as a network of networks — millions of LANs, MANs and WANs agreeing to use the same rules so that any host can reach any other host.

3. Network topologies

The **topology** is the shape in which nodes are connected. The shape decides cost, ease of adding machines, and what happens when a cable or a central device fails.



Four basic shapes: one shared backbone, one centre, one loop, many cross links.

Figure 1 — Bus, star, ring and mesh topologies

TOPOLOGY	ADVANTAGES	DISADVANTAGES
Bus	Cheap; uses the least cable; easy to add a node	A break in the main cable stops the whole network; collisions grow as traffic grows

Star	One failed cable affects only one node; easy to find faults; the common choice today	If the central switch fails, the whole network fails; needs more cable
Ring	Orderly access, no collisions in token designs; equal treatment of nodes	One break can stop the ring; adding a node interrupts the network
Mesh	Very reliable — many alternative paths; a failed link rarely isolates a node	Most expensive; cabling and ports grow very fast with the number of nodes

Modern office LANs are almost always wired as a star around a switch, even when the internal working of the switch makes the traffic behave in a more complex way. The topology describes the physical or logical arrangement, and exam answers should say which one is meant.

4. Network devices

DEVICE	WHAT IT DOES
NIC — Network Interface Card	The adapter inside (or attached to) each computer that connects it to the network. Every NIC carries a unique hardware address, the MAC address, used inside a local network.
Hub	A simple central connector: it repeats every incoming signal to all ports. Cheap, but wasteful and insecure, because every node receives every message.
Switch	A smarter central connector: it learns which node sits on which port and forwards each frame only to its destination. The normal centre of a modern LAN.
Router	Connects different networks together and chooses the path packets should take between them. Your home router joins your LAN to your Internet provider.
Modem	Converts between the digital signals of a computer and the form carried by a telephone, cable or fibre line — modulation and demodulation, hence the name.
Gateway	A node that joins two networks using different protocols and translates between them, so that unlike systems can still exchange data.

The hub-versus-switch contrast is a favourite exam point: a hub *broadcasts* to everyone, a switch *forwards* to the intended recipient only. A router works one level higher, between networks rather than inside one network.

5. Transmission media

Data needs a physical path. **Twisted pair** cable — pairs of copper wires twisted together to reduce interference — is the standard LAN cable (the familiar Ethernet cable with an RJ45 plug); it is cheap

and easy to install, with common lengths up to about 100 m per run. **Coaxial cable**, a central copper core inside a metal shield, was common in older networks and is still used for cable television feeds; it resists interference better than unshielded twisted pair but is stiffer and costlier. **Fibre optic** cable carries pulses of light through glass or plastic strands: it gives the highest speeds and longest distances, is immune to electrical interference, and is the backbone medium of the Internet and of undersea cables, at a higher price and with more delicate handling.

Wireless media use radio waves instead of a cable: Wi-Fi inside buildings, mobile networks (4G, 5G) across cities, Bluetooth over a few metres, and satellite links where nothing else reaches. Wireless is convenient and supports moving users, but it is generally slower than a good cable, is shared with nearby users, and must be protected carefully because its signals leave the room.

6. Client–server and peer-to-peer models

In the **client–server** model, powerful central computers called **servers** provide services — files, web pages, email, databases — and the users' computers, the **clients**, request them. Control, security and backup are centralised, which suits a university or a company; the cost is that the server is expensive and its failure affects everyone. In the **peer-to-peer** model, all computers are equal peers: each can both request and offer files or services. It is cheap and simple for a small home group, but hard to secure, back up or administer as numbers grow. Most real systems mix the two: the public Internet looks peer-like in design, while almost every service you actually use — mail, banking, learning platforms — is client–server in practice.

7. The Internet: a brief history

The Internet grew from **ARPANET**, a United States research network that connected its first four university nodes in 1969. Its key idea, packet switching — cutting messages into small packets that travel independently and are reassembled at the destination — made networks robust and efficient. In 1983 ARPANET adopted the TCP/IP protocol suite as its standard, a date often taken as the birth of the modern Internet. Through the 1980s academic and research networks joined; in 1989–1991 Tim Berners-Lee at CERN invented the World Wide Web — pages, links and the first browser — and in the 1990s commercial providers opened the network to the public. The result is not owned by any single company or government: it is a cooperative system held together by shared technical standards and agreements between the networks that compose it.

8. Addresses, names and the Web

Every host on the Internet needs an address. An **IPv4 address** is a 32-bit number written in **dotted decimal** form: four decimal numbers from 0 to 255 separated by dots, for example 192.168.0.1. Each

of the four parts is one byte (8 bits), so 4 × 8 = 32 bits in total, giving about 4.3 billion possible addresses — a number that once seemed endless and is now effectively exhausted, which is why the much larger IPv6 system is being adopted alongside it. Historically IPv4 addresses were grouped into classes (A, B, C, D, E) by the size of network they served; modern practice uses flexible subnetting instead, so classes are now mainly background knowledge.

Numbers suit machines, not people, so the **Domain Name System (DNS)** translates names such as `www.example.com` into IP addresses automatically, the way a phone book turns a name into a number. When you type a name, your computer asks a DNS server for the address, then connects to that address; if DNS fails, the site appears "down" even though the server itself is healthy.

The **World Wide Web (WWW)** is not the same thing as the Internet. The Internet is the underlying network — the cables, routers and protocols. The Web is *one service* that runs on top of it: a system of interlinked pages and resources identified by URLs and transferred with HTTP. Email, file transfer and video calls also run on the Internet without being part of the Web. A **web browser** (the client program on your device) requests pages; a **web server** stores the site and sends the pages back. Each resource has a **Uniform Resource Locator (URL)**, whose parts can be read directly:

https://www.example.com:443/courses/index.html

Here `https` is the protocol (scheme), `www.example.com` is the host name that DNS resolves, `443` is the port (the standard HTTPS port, usually omitted in writing), and `/courses/index.html` is the path of the requested file on that server.

9. Common protocols

A **protocol** is an agreed set of rules for communication: how a conversation starts, how data is formatted, how errors are handled and how it ends. Without shared protocols, machines from different makers could not understand each other at all.

PROTOCOL	JOB
TCP — Transmission Control Protocol	Breaks data into packets, numbers them, checks arrival, resends what is lost and reassembles everything in order. Reliable delivery.
IP — Internet Protocol	Addresses each packet and routes it across networks towards its destination address. TCP and IP work as a pair: TCP/IP.
HTTP / HTTPS	Transfers web pages between server and browser. HTTPS is the same transfer protected by encryption, and is the expected default today.
FTP — File Transfer Protocol	Transfers files between a client and a server, with upload and download operations.

SMTP — Simple Mail Transfer Protocol	Sends outgoing email and passes mail between mail servers.
POP3 / IMAP	Retrieve incoming email from a mail server. POP3 traditionally downloads mail to one device; IMAP keeps mail on the server and synchronises it across devices.
DNS — Domain Name System	Translates domain names into IP addresses, as described in Section 8.

10. Internet services

On top of this machinery sit the services people actually name. **Email** remains the formal written channel of business and education. **Electronic commerce (e-commerce)** — buying and selling goods and services online, with online payment — now covers everything from a single artisan shop to national marketplaces. **Cloud computing** rents computing as a service: storage, software and processing run on a provider's servers and are reached through the network, so a student can keep files in cloud storage, edit them in a browser-based office suite, and open them from any device. Other everyday services include search engines, social networking, video streaming, Internet banking, online education platforms and voice and video calling. In a textile setting the same services carry buyer orders, design approvals, shipment tracking and video inspections between a factory in Bangladesh and a buyer on another continent.

WORKED EXAMPLE

How long does a download take? A file of 50 MB is transferred over a link rated at 10 Mbps (megabits per second). File sizes are measured in bytes and link speeds in bits, and 1 byte = 8 bits, so first convert the file: 50 MB × 8 = 400 Mbit. Time = amount ÷ rate = 400 Mbit ÷ 10 Mbps = **40 s**.

Assumptions: the link delivers its full rated speed for the whole transfer, the file size uses the simple decimal convention of this estimate, and no allowance is made for protocol overhead, other users sharing the link, or time to start the transfer. Real downloads are therefore usually a little slower than this ideal figure.

11. Network security basics

Connecting a machine exposes it, so a few defences are considered basic hygiene. **Passwords** should be long, unique to each account, and never shared; where offered, a second verification step should be turned on. A **firewall** — software, or a device, sitting between a network and the outside — inspects incoming and outgoing traffic and blocks connections that break its rules. **Encryption** scrambles data

so that only the intended receiver can read it; it protects web traffic (HTTPS), messages and stored files even if the data is intercepted on the way.

The common threats are grouped under the name **malware** (malicious software). A **virus** attaches itself to files or programs and spreads when they are shared. A **worm** spreads by itself across a network without needing a host file. A **Trojan horse** pretends to be useful software while hiding a harmful action. **Ransomware** encrypts the victim's files and demands payment for the key. Beyond software, **phishing** attacks the person: a fake message or page imitating a trusted body, designed to steal passwords or money. Updated software, regular backups, a working firewall and a suspicious eye on unexpected messages together stop most real attacks.

Glossary

TERM	MEANING
Node	Any device connected to a network: a computer, printer, phone or server.
Topology	The shape — bus, star, ring, mesh and others — in which the nodes of a network are connected.
Switch	A central device that forwards each message only to the port of its intended recipient.
Router	A device that connects different networks and chooses the path packets take between them.
Bandwidth	The maximum data rate of a link, usually measured in bits per second.
IP address	The numeric address identifying a host on an IP network; IPv4 uses 32 bits in dotted decimal form.
DNS	The Domain Name System: the service that translates domain names into IP addresses.
Protocol	An agreed set of rules that governs how two systems communicate.
Firewall	Software or hardware that inspects network traffic and blocks connections that break its rules.
Malware	Malicious software as a group: viruses, worms, Trojan horses, ransomware and similar threats.

Key points & summary

- A network connects nodes to share resources, files and communication, with central control and backup as further benefits.
- LAN covers a building, MAN a city, WAN a country or the world; the Internet is a network of networks.
- Star is the practical default topology today; a switch forwards to one destination where a hub broadcasts to all.
- Fibre optic media give the highest speed and distance; wireless trades some speed and safety for mobility.
- IPv4 addresses are 32 bits in dotted decimal; DNS turns names into addresses; the Web is one service running on the Internet, not the Internet itself.
- TCP gives reliable, ordered delivery while IP addresses and routes the packets; HTTPS is HTTP protected by encryption.
- Transfer time = file size in bits ÷ link rate: 50 MB over 10 Mbps takes 40 s under ideal assumptions.

Practice

Answer in your own words; brief model answers follow.

1. State three benefits of connecting the computers of a small office into a LAN, and one risk that the connection creates.
2. Distinguish between a LAN, a MAN and a WAN by span and by one example of each.
3. A school wants a topology that keeps working for other rooms when one classroom cable is cut, but cannot afford a full mesh. Which topology fits best, and why?
4. Explain, in two or three sentences each, the different jobs of a switch and a router.
5. A user types a website name and the page cannot open, although the same site opens when its numeric address is used. Which service has most likely failed, and what does that service normally do?
6. How long would a 100 MB file take over a 20 Mbps link, under the same ideal assumptions used in this sheet? Show the conversion step.

Answer key

1. Any three of: sharing one printer and one Internet connection, sharing files in their latest version, central backup and account control, easy internal communication. Risk: an attack or malware can now reach machines through the network, so security must be managed.
2. LAN: a building or small campus, e.g. a university laboratory. MAN: a city, e.g. a cable network or linked city campuses. WAN: a country or the world, e.g. the Internet or a bank’s national branch

network.

3. Star. Each room has its own cable to a central switch, so cutting one cable disconnects only that room; a mesh would need a cable between every pair of rooms and costs far more.
4. A switch works inside one network: it learns which node is on which port and forwards each frame only to its destination. A router works between networks: it reads the destination address and chooses the path that carries the packet towards the correct network.
5. DNS, the Domain Name System. It normally translates the typed domain name into the site’s IP address; the direct numeric address works because that translation step is then not needed.
6. 100 MB × 8 = 800 Mbit. Time = 800 ÷ 20 = 40 s.
